

DOC325 Digital Circuits and Logic Design

Quiz Questions Summary

Quiz Review

September 2022

Questions and Selected Answers

Question 1: What is the decimal equivalent of binary number 1000010:

- **Selected Answer:** c. 66
- *Logic:* $1 \cdot 2^6 + 0 \cdot 2^5 + 0 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 = 64 + 2 = 66.$

Question 2: Logically, an EXCLUSIVE NOR gate is equal to:

- **Selected Answer:** b. An exclusive OR gate followed by an inverter.

Question 3: Find LSD and MSD in the following decimal number: 0000000099

- **LSD (Least Significant Digit):** 9
- **MSD (Most Significant Digit):** 9

Question 4: The result of 01101101101 - 1001 - 111111 in binary is:

- **Selected Answer:** d. 1100100101

Question 5: An AND gate can be imagined as:

- **Selected Answer:** a. switches connected in series.

Question 6: Which logic gate is depicted by the following truth table?

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

- **Selected Answer:** d. XOR

Question 7: Logic expressions in the circuit (Question 7):

- First gate (Top AND): AB
- Second gate (Middle AND): ABC'
- Third gate (Bottom OR): $B + C$
- Fourth gate (Bottom AND): $B(B + C)$
- Final Output (Lamp): $AB + ABC' + B(B + C)$

Question 8: The multiplication of the following binary numbers 11111 X 10001 are:

- **Selected Answer:** a. 1000001111
- *Logic:* $31 \times 17 = 527$. Converting 527 to binary gives 1000001111_2 .

Question 9: The hexadecimal equivalent of the binary number 1010011010101011 is:

- **Selected Answer:** d. A6AB
- *Logic:* Grouping into nibbles: $(1010)_2 = A_{16}$, $(0110)_2 = 6_{16}$, $(1010)_2 = A_{16}$, $(1011)_2 = B_{16}$.

Question 11: The decimal equivalent of the hexadecimal number 62 is:

- **Selected Answer:** c. 98
- *Logic:* $(6 \times 16^1) + (2 \times 16^0) = 96 + 2 = 98$.

Question 12: The sum of the 11011000000 + 11100111 + 11101011 in binary is:

- **Selected Answer:** a. 100010010010

Question 13: The 2's complement of the binary number 10101010 is:

- **Selected Answer:** a. 01010110
- *Logic:* 1's complement = 01010101. Adding 1 = 01010110.

Question 15: The function of NOT is to:

- **Selected Answer:** a. Invert input signal.

Question 16: Which logic gate is depicted by the following truth table?

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	1

- **Selected Answer:** a. XNOR

Question 17: The binary equivalent of the hexadecimal number E2F is:

- **Selected Answer:** d. 111000101111
- *Logic:* $E = 1110$, $2 = 0010$, $F = 1111$.

Question 18: The binary equivalent of the decimal number 255 is:

- **Selected Answer:** c. 11111111

Question 19: The hexadecimal equivalent of the decimal number 971 is:

- **Correct Answer:** a. 3CB
- *Logic:* $971 \div 16 = 60 \text{ rem } 11(B)$; $60 \div 16 = 3 \text{ rem } 12(C)$; $3 \div 16 = 0 \text{ rem } 3$. Reading remainders bottom-up: 3CB.